

Rule base of the deployed fuzzy inference system

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Mamdani inference, 42 rules, single-ear mode. Extracted from the built control system (fuzzy_audiogram.rules), not transcribed, and reconciled against the 42 rules the assembled FIS actually contains.

The source defines the configuration rules with one if severity is not None / else pair, so counting ctrl.Rule call sites gives 43. In single-ear mode the severity-free branch is the one taken, giving 14 configuration rules and 42 in total. That is the figure the manuscript states.

Antecedents are joined by AND. Each variable is a linguistic term of the fuzzified input named.

A. Severity rules (12)

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#	Antecedent	Consequent
1	threshold is normal	severity = normal
2	threshold is mild	severity = mild
3	threshold is moderate	severity = moderate
4	threshold is moderately_severe	severity = moderately_severe
5	threshold is severe	severity = severe
6	threshold is profound	severity = profound
7	threshold is normal and threshold is mild	severity = mild
8	threshold is mild and threshold is moderate	severity = moderate
9	threshold is moderate and threshold is moderately_severe	severity = moderately_severe
10	threshold is moderately_severe and threshold is severe	severity = severe
11	threshold is severe and threshold is profound	severity = severe
12	threshold is normal and threshold is mild and threshold is moderate	severity = moderate

B. Configuration rules (14)

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#	Antecedent	Consequent
13	slope is flat and notch is no_notch	audiogram_shape = flat
14	slope is gently_sloping and notch is no_notch	audiogram_shape = sloping
15	slope is steeply_sloping and notch is no_notch	audiogram_shape = sloping
16	slope is precipitous and notch is no_notch	audiogram_shape = precipitous
17	slope is rising and notch is no_notch	audiogram_shape = rising
18	slope is flat and notch is no_notch	audiogram_shape = normal
19	notch is shallow_notch and slope is flat	audiogram_shape = notched
20	notch is deep_notch and slope is flat	audiogram_shape = notched
21	notch is shallow_notch and slope is gently_sloping	audiogram_shape = notched

22	notch is deep_notch and slope is steeply_sloping	audiogram_shape = notched
23	slope is rising and notch is deep_notch	audiogram_shape = rising
24	slope is precipitous and notch is shallow_notch	audiogram_shape = precipitous
25	slope is gently_sloping and notch is deep_notch	audiogram_shape = notched
26	slope is steeply_sloping and notch is deep_notch	audiogram_shape = precipitous

C. Complex configuration interaction rules (4)

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#	Antecedent	Consequent
27	slope is precipitous and threshold is moderate	severity = severe
28	slope is precipitous and threshold is mild	severity = moderately_severe
29	threshold is normal and slope is steeply_sloping	severity = mild
30	threshold is normal and slope is precipitous	severity = moderate

D. Referral rules (12)

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#	Antecedent	Consequent
31	asymmetry is symmetric	referral = none
32	asymmetry is mildly_asymmetric	referral = routine
33	asymmetry is moderately_asymmetric	referral = expedited
34	asymmetry is severely_asymmetric	referral = urgent
35	asymmetry is mildly_asymmetric and referral is none	referral = routine
36	asymmetry is moderately_asymmetric and referral is routine	referral = expedited
37	asymmetry is severely_asymmetric and referral is expedited	referral = urgent
38	asymmetry is mildly_asymmetric and referral is routine	referral = routine
39	asymmetry is moderately_asymmetric and referral is expedited	referral = expedited
40	asymmetry is severely_asymmetric and referral is urgent	referral = urgent
41	asymmetry is severely_asymmetric	referral = urgent
42	asymmetry is moderately_asymmetric	referral = expedited

Total: 42 rules — 12 severity, 14 configuration, 4 complex interaction, 12 referral.

Verified against the assembled FIS, which contains 42 rules.